

Домашняя работа №3

Анализ трафика в Wireshark

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1. Цели и задачи работы
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1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Изучение MAC-адресации, анализ кадров канального уровня, а также исследование протоколов ARP, ICMP, HTTP, DNS, TCP, UDP и процесса TCP handshake с помощью Wireshark.

2. 2. Процесс выполнения лабораторной работы

2.1 Просматриваю сетевую конфигурацию

Адаптер Ethernet Ethernet 2:

```
DNS-суффикс подключения . . . . . :  
Локальный IPv6-адрес канала . . . : fe80::c080:3353:3af3:6fc8%18  
IPv4-адрес. . . . . : 192.168.56.1  
Маска подсети . . . . . : 255.255.255.0  
Основной шлюз. . . . . :
```

Адаптер Ethernet Ethernet 3:

```
DNS-суффикс подключения . . . . . :  
Локальный IPv6-адрес канала . . . : fe80::2658:cf7e:27cd:6a8a%55  
IPv4-адрес. . . . . : 10.21.58.50  
Маска подсети . . . . . : 255.255.255.0  
Основной шлюз. . . . . : 10.21.58.106
```

Рисунок 1: Просмотр основной сетевой конфигурации с помощью команды ipconfig

2.2 Определяю MAC-адреса интерфейсов

```
Адаптер Ethernet Ethernet 2:

DNS-суффикс подключения . . . . . : 
Описание. . . . . : VirtualBox Host-Only Ethernet Adapter
Физический адрес. . . . . : 0A-00-27-00-00-12
DHCP включен. . . . . : Нет
Автонастройка включена. . . . . : Да
Локальный IPv6-адрес канала . . . : fe80::c080:3353:3af3:6fc8%18(Основной)
IPv4-адрес. . . . . : 192.168.56.1(Основной)
Маска подсети . . . . . : 255.255.255.0
Основной шлюз. . . . . : 
IAID DHCPv6 . . . . . : 453640231
DUID клиента DHCPv6 . . . . . : 00-01-00-01-30-47-94-1A-10-FF-E0-21-D6-B4
NetBios через TCP/IP. . . . . : Включен

Адаптер Ethernet Ethernet 3:

DNS-суффикс подключения . . . . . : 
Описание. . . . . : Remote NDIS Compatible Device
Физический адрес. . . . . : 12-A3-EF-62-D1-5B
DHCP включен. . . . . : Да
Автонастройка включена. . . . . : Да
Локальный IPv6-адрес канала . . . : fe80::2658:cf7e:27cd:6a8a%55(Основной)
IPv4-адрес. . . . . : 10.21.58.50(Основной)
Маска подсети . . . . . : 255.255.255.0
Аренда получена. . . . . : 26 сентября 2026 г. 12:27:20
Срок аренды истекает. . . . . : 26 сентября 2026 г. 13:57:17
Основной шлюз. . . . . : 10.21.58.106
DHCP-сервер. . . . . : 10.21.58.106
IAID DHCPv6 . . . . . : 923968495
DUID клиента DHCPv6 . . . . . : 00-01-00-01-30-47-94-1A-10-FF-E0-21-D6-B4
DNS-серверы. . . . . : 10.21.58.106
NetBios через TCP/IP. . . . . : Включен
```

Рисунок 2: Получение расширенной сетевой конфигурации и MAC-адресов с помощью `ipconfig /all`

2.3 Проверяю доступность шлюза

```
PS D:\work\paulinelourdes\vagrant> ping 10.21.58.106

Обмен пакетами с 10.21.58.106 по с 32 байтами данных:
Ответ от 10.21.58.106: число байт=32 время=1мс TTL=64
Ответ от 10.21.58.106: число байт=32 время<1мс TTL=64
Ответ от 10.21.58.106: число байт=32 время<1мс TTL=64
Ответ от 10.21.58.106: число байт=32 время<1мс TTL=64

Статистика Ping для 10.21.58.106:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)

Приблизительное время приема-передачи в мс:
    Минимальное = 0мсек, Максимальное = 1 мсек, Среднее = 0 мсек
PS D:\work\paulinelourdes\vagrant> |
```

Рисунок 3: Проверка доступности шлюза по умолчанию с помощью ping

2.4 Анализирую ICMP Echo Request

The image shows a Wireshark packet capture analysis of an ICMP Echo Request. The top pane displays a list of packets, with packet 34 selected. The middle pane shows the packet details for packet 34, and the bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
27	7.111230	3a:a3:2b:ce:b3:84	12:a3:ef:62:d1:5b	ARP	42	Who has 10.21.58.50? Tell 10.21.58.106
28	7.111250	12:a3:ef:62:d1:5b	3a:a3:2b:ce:b3:84	ARP	42	10.21.58.50 is at 12:a3:ef:62:d1:5b
34	9.412671	10.21.58.50	10.21.58.106	ICMP	74	Echo (ping) request id=0x0001, seq=17/4352, ttl=128 (reply in 35)
35	9.413778	10.21.58.106	10.21.58.50	ICMP	74	Echo (ping) reply id=0x0001, seq=17/4352, ttl=64 (request in 34)

Packet Details (Frame 34):

- Frame 34: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{234AD...}
- Ethernet II, Src: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b), Dst: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)
 - Destination: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)
 - ...1. = LG bit: Locally administered address (this is NOT the factory default)
 - ...0 = IG bit: Individual address (unicast)
 - Source: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)
 - ...1. = LG bit: Locally administered address (this is NOT the factory default)
 - ...0 = IG bit: Individual address (unicast)
 - Type: IPv4 (0x0800)
 - [Stream index: 0]
- Internet Protocol Version 4, Src: 10.21.58.50, Dst: 10.21.58.106
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 60
 - Identification: 0x9bb4 (39860)
 - > 000. = Flags: 0x0
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 128
 - Protocol: ICMP (1)
 - Header Checksum: 0x1647 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 10.21.58.50
 - Destination Address: 10.21.58.106
 - [Stream index: 5]
- Internet Control Message Protocol
 - Type: Echo (ping) request (8)
 - Code: 0
 - Checksum: 0x4d4a [correct]
 - [Checksum Status: Good]
 - Identifier (BE): 1 (0x0001)
 - Identifier (LE): 256 (0x0100)
 - Sequence Number (BE): 17 (0x0011)
 - Sequence Number (LE): 4352 (0x1100)
 - [Response frame: 35]
 - > Data (32 bytes)

Raw Data (Hex):

```
0000 3a a3 2b ce b3 84 12 a3 ef 62 d1 5b 08 00 45 00
0010 00 3c 9b b4 00 00 00 01 16 47 0a 15 3a 32 0a 15
0020 3a 6a 08 00 4d 4a 00 01 00 11 61 62 63 64 65 66
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
0040 77 61 62 63 64 65 66 67 68 69
```

Рисунок 4: Анализ ICMP Echo Request к шлюзу в Wireshark

2.5 Анализирую ICMP Echo Reply

The image shows a Wireshark packet capture analysis of an ICMP Echo Reply. The top pane displays a list of packets, with packet 35 selected. The middle pane shows the packet details for the selected packet, and the bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
27	7.111230	3a:a3:2b:ce:b3:84	12:a3:ef:62:d1:5b	ARP	42	Who has 10.21.58.50? Tell 10.21.58.106
28	7.111250	12:a3:ef:62:d1:5b	3a:a3:2b:ce:b3:84	ARP	42	10.21.58.50 is at 12:a3:ef:62:d1:5b
34	9.412671	10.21.58.50	10.21.58.106	ICMP	74	Echo (ping) request id=0x0001, seq=17/4352, ttl=128 (reply in 35)
35	9.413778	10.21.58.106	10.21.58.50	ICMP	74	Echo (ping) reply id=0x0001, seq=17/4352, ttl=64 (request in 34)

Packet Details:

- Ethernet II, Src:** 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84), **Dst:** 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)
 - Destination:** 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)
 - ...1... = LG bit: Locally administered address (this is NOT the factory default)
 - ...0... = IG bit: Individual address (unicast)
 - Source:** 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)
 - ...1... = LG bit: Locally administered address (this is NOT the factory default)
 - ...0... = IG bit: Individual address (unicast)
 - Type: IPv4 (0x0800)
 - [Stream index: 0]
- Internet Protocol Version 4, Src:** 10.21.58.106, **Dst:** 10.21.58.50
 - 0100 ... = Version: 4
 - ...0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field:** 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 60
 - Identification: 0x09a0 (2464)
 - Flags:** 0x00
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 64
 - Protocol: ICMP (1)
 - Header Checksum: 0xe85b [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 10.21.58.106
 - Destination Address: 10.21.58.50
 - [Stream index: 5]
- Internet Control Message Protocol**
 - Type: Echo (ping) reply (0)
 - Code: 0
 - Checksum: 0x554a [correct]
 - [Checksum Status: Good]
 - Identifier (BE): 1 (0x0001)
 - Identifier (LE): 256 (0x0100)
 - Sequence Number (BE): 17 (0x0011)
 - Sequence Number (LE): 4352 (0x1100)
 - [Request from: 34]
 - [Response time: 1,107 ms]
 - Data (32 bytes)

Raw Data:

```
0000 12 a3 ef 62 d1 5b 3a a3 2b ce b3 84 08 00 45 00
0010 00 3c 09 a0 00 00 40 01 e8 5b 0a 15 3e 6a 0a 15
0020 3a 32 00 00 55 4a 00 01 00 11 61 62 63 64 65 66
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
0040 77 61 62 63 64 65 66 67 68 69
```

Рисунок 5: Анализ ICMP Echo Reply от шлюза в Wireshark

2.6 Изучаю ARP-запрос

arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
27	7.111230	3a:a3:2b:ce:b3:84	12:a3:ef:62:d1:5b	ARP	42	Who has 10.21.58.50? Tell 10.21.58.106
28	7.111250	12:a3:ef:62:d1:5b	3a:a3:2b:ce:b3:84	ARP	42	10.21.58.50 is at 12:a3:ef:62:d1:5b
34	9.412671	10.21.58.50	10.21.58.106	ICMP	74	Echo (ping) request id=0x0001, seq=17/4352, ttl=128 (reply in 35)
35	9.413778	10.21.58.106	10.21.58.50	ICMP	74	Echo (ping) reply id=0x0001, seq=17/4352, ttl=64 (request in 34)

> Frame 27: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{234AD...}

▼ Ethernet II, Src: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84), Dst: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)

Destination: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)

...1. = LG bit: Locally administered address (this is NOT the factory default)

...0. = IG bit: Individual address (unicast)

Source: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)

...1. = LG bit: Locally administered address (this is NOT the factory default)

...0. = IG bit: Individual address (unicast)

Type: ARP (0x0806)

[Stream index: 0]

▼ Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)

Sender IP address: 10.21.58.106

Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)

Target IP address: 10.21.58.50

0000 12 a3 ef 62 d1 5b 3a a3 2b ce b3 84 06 00 01

0010 08 00 06 04 00 01 3a a3 2b ce b3 84 0a 15 3a 6a

0020 00 00 00 00 00 00 0a 15 3a 32

Рисунок 6: Анализ ARP-запроса в Wireshark

2.7 Изучаю ARP-ответ

arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
27	7.111230	3a:a3:2b:ce:b3:84	12:a3:ef:62:d1:5b	ARP	42	Who has 10.21.58.50? Tell 10.21.58.106
28	7.111250	12:a3:ef:62:d1:5b	3a:a3:2b:ce:b3:84	ARP	42	10.21.58.50 is at 12:a3:ef:62:d1:5b
34	9.412671	10.21.58.50	10.21.58.106	ICMP	74	Echo (ping) request id=0x0001, seq=17/4352, ttl=128 (reply in 35)
35	9.413778	10.21.58.106	10.21.58.50	ICMP	74	Echo (ping) reply id=0x0001, seq=17/4352, ttl=64 (request in 34)

> Frame 28: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{234AD...}	0000	3a a3 2b ce b3 84 12 a3 ef 62 d1 5b 08 06 00 01
▼ Ethernet II, Src: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b), Dst: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)	0010	08 00 06 04 00 02 12 a3 ef 62 d1 5b 0a 15 3a 32
▼ Destination: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)	0020	3a a3 2b ce b3 84 0a 15 3a 6a
....1. = LG bit: Locally administered address (this is NOT the factory default)		
....0. = IG bit: Individual address (unicast)		
▼ Source: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)		
....1. = LG bit: Locally administered address (this is NOT the factory default)		
....0. = IG bit: Individual address (unicast)		
Type: ARP (0x0806)		
[Stream index: 0]		
▼ Address Resolution Protocol (reply)		
Hardware type: Ethernet (1)		
Protocol type: IPv4 (0x0800)		
Hardware size: 6		
Protocol size: 4		
Opcode: reply (2)		
Sender MAC address: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)		
Sender IP address: 10.21.58.50		
Target MAC address: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)		
Target IP address: 10.21.58.106		

Рисунок 7: Анализ ARP-ответа в Wireshark

2.8 Проверяю доступность удалённого узла

```
PS D:\work\paulinelourdes\vagrant> ping yandex.ru

Обмен пакетами с yandex.ru [5.255.255.77] с 32 байтами данных:
Ответ от 5.255.255.77: число байт=32 время=19мс TTL=50
Ответ от 5.255.255.77: число байт=32 время=39мс TTL=50
Ответ от 5.255.255.77: число байт=32 время=27мс TTL=50
Ответ от 5.255.255.77: число байт=32 время=23мс TTL=50

Статистика Ping для 5.255.255.77:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)

Приблизительное время приема-передачи в мс:
    Минимальное = 19мсек, Максимальное = 39 мсек, Среднее = 27 мсек
PS D:\work\paulinelourdes\vagrant> |
```

Рисунок 8: Проверка доступности yandex.ru с помощью команды ping

2.9 Анализирую ICMP-запрос к удалённому узлу

arp or icmp					
No.	Time	Source	Destination	Protocol	Length Info
56	9.198089	10.21.58.50	5.255.255.77	ICMP	74 Echo (ping) request id=0x0001, seq=21/5376, ttl=128 (reply in 57)
57	9.217939	5.255.255.77	10.21.58.50	ICMP	74 Echo (ping) reply id=0x0001, seq=21/5376, ttl=50 (request in 56)
58	10.201438	10.21.58.50	5.255.255.77	ICMP	74 Echo (ping) request id=0x0001, seq=22/5632, ttl=128 (reply in 61)
61	10.240858	5.255.255.77	10.21.58.50	ICMP	74 Echo (ping) reply id=0x0001, seq=22/5632, ttl=50 (request in 58)
69	11.214113	10.21.58.50	5.255.255.77	ICMP	74 Echo (ping) request id=0x0001, seq=23/5888, ttl=128 (reply in 70)
70	11.241930	5.255.255.77	10.21.58.50	ICMP	74 Echo (ping) reply id=0x0001, seq=23/5888, ttl=50 (request in 69)
71	12.228305	10.21.58.50	5.255.255.77	ICMP	74 Echo (ping) request id=0x0001, seq=24/6144, ttl=128 (reply in 72)
72	12.251761	5.255.255.77	10.21.58.50	ICMP	74 Echo (ping) reply id=0x0001, seq=24/6144, ttl=50 (request in 71)
256	24.067262	3a:a3:2b:ce:b3:84	12:a3:ef:62:d1:5b	ARP	42 Who has 10.21.58.50? Tell 10.21.58.106

> Frame 56: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{234AD...}	0000	3a a3 2b ce b3 84 12 a3 ef 62 d1 5b 00 00 45 00	:a.....
↳ Ethernet II, Src: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b), Dst: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)	0010	00 3c 62 69 00 00 00 01 8e c4 0a 15 3a 32 05 ff	cbi....
↳ Destination: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)	0020	ff 4d 08 00 4d 46 00 01 00 15 61 62 63 64 65 66	M..MF..
↳ Source: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)	0030	67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76	ghijklm
↳ Source: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)	0040	77 61 62 63 64 65 66 67 68 69	wabcedfg
Type: IPv4 (0x0800)			
[Stream index: 0]			
↳ Internet Protocol Version 4, Src: 10.21.58.50, Dst: 5.255.255.77			
0100 = Version: 4			
... 0101 = Header Length: 20 bytes (5)			
↳ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)			
Total Length: 60			
Identification: 0x6269 (25193)			
0000 = Flags: 0x0			
... 0 0000 0000 0000 = Fragment Offset: 0			
Time to Live: 128			
Protocol: ICMP (1)			
Header Checksum: 0xBec4 [validation disabled]			
[Header checksum status: Unverified]			
Source Address: 10.21.58.50			
Destination Address: 5.255.255.77			
[Stream index: 6]			
↳ Internet Control Message Protocol			
Type: Echo (ping) request (8)			
Code: 0			
Checksum: 0x4d46 [correct]			
[Checksum Status: Good]			
Identifier (BE): 1 (0x0001)			
Identifier (LE): 256 (0x0100)			
Sequence Number (BE): 21 (0x0015)			
Sequence Number (LE): 5376 (0x1500)			
↳ Response frame 57			
Data (32 bytes)			

Рисунок 9: Анализ ICMP Echo Request к удалённому узлу

2.10 Анализирую ICMP-ответ от удалённого узла

arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
56	9.198889	10.21.58.50	5.255.255.77	ICMP	74	Echo (ping) request id=0x0001, seq=21/5376, ttl=128 (reply in 57)
57	9.217939	5.255.255.77	10.21.58.50	ICMP	74	Echo (ping) reply id=0x0001, seq=21/5376, ttl=50 (request in 56)
58	10.201438	10.21.58.50	5.255.255.77	ICMP	74	Echo (ping) request id=0x0001, seq=22/5632, ttl=128 (reply in 61)
61	10.240858	5.255.255.77	10.21.58.50	ICMP	74	Echo (ping) reply id=0x0001, seq=22/5632, ttl=50 (request in 58)
69	11.214113	10.21.58.50	5.255.255.77	ICMP	74	Echo (ping) request id=0x0001, seq=23/5888, ttl=128 (reply in 70)
70	11.241930	5.255.255.77	10.21.58.50	ICMP	74	Echo (ping) reply id=0x0001, seq=23/5888, ttl=50 (request in 69)
71	12.228305	10.21.58.50	5.255.255.77	ICMP	74	Echo (ping) request id=0x0001, seq=24/6144, ttl=128 (reply in 72)
72	12.251761	5.255.255.77	10.21.58.50	ICMP	74	Echo (ping) reply id=0x0001, seq=24/6144, ttl=50 (request in 71)
256	24.067262	3a:a3:2b:ce:b3:84	12:a3:ef:62:d1:5b	ARP	42	who has 10.21.58.50? Tell 10.21.58.106

> Frame 57: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{234}	0000	12 a3 ef 62 d1 5b 3a a3 2b ce b3 84 00 00 45 28	...b [1-
▼ Ethernet II, Src: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84), Dst: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)	0010	00 3c 62 69 00 32 01 dc 9c 05 ff ff 4d 8a 15	...b1-2-
▼ Destination: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)	0020	3a 32 00 00 55 46 00 01 00 15 61 62 63 64 65 66	12-UF...
....0 = LG bit: Locally administered address (this is NOT the factory default)	0030	67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76	ghijklm
▼ Source: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)	0040	77 61 62 63 64 65 66 67 68 69	wabdefg
....1 = LG bit: Locally administered address (this is NOT the factory default)			
....0 = LG bit: Individual address (unicast)			
Type: IPv4 (0x0800)			
[Stream index: 0]			
▼ Internet Protocol Version 4, Src: 5.255.255.77, Dst: 10.21.58.50			
0100 = Version: 4			
....0101 = Header Length: 20 bytes (5)			
> Differentiated Services Field: 0x28 (DSCP: AF11, ECN: Not-ECT)			
Total Length: 60			
Identification: 0x6269 (25193)			
> 000. Flags: 0x0			
...0 0000 0000 0000 = Fragment Offset: 0			
Time to Live: 50			
Protocol: ICMP (1)			
Header Checksum: 0xdc9c [validation disabled]			
[Header checksum status: Unverified]			
Source Address: 5.255.255.77			
Destination Address: 10.21.58.50			
[Stream index: 6]			
▼ Internet Control Message Protocol			
Type: Echo (ping) reply (0)			
Code: 0			
Checksum: 0x5546 [correct]			
[Checksum Status: Good]			
Identifier (BE): 1 (0x0001)			
Identifier (LE): 256 (0x0100)			
Sequence Number (BE): 21 (0x0015)			
Sequence Number (LE): 5376 (0x1500)			
[Request frame: 56]			
[Response time: 10,850 ms]			
> Data (42 bytes)			

Рисунок 10: Анализ ICMP Echo Reply от удалённого узла

2.11 Исследуя HTTP-запрос

No.	Time	Source	Destination	Protocol	Length	Info
329	5.968346	10.21.58.50	188.184.67.127	HTTP	342	GET /apple-touch-icon.png HTTP/1.1
332	6.053248	188.184.67.127	10.21.58.50	HTTP	414	HTTP/1.1 404 Not Found (text/html)
357	6.532674	10.21.58.50	188.184.67.127	HTTP	497	GET / HTTP/1.1
359	6.634118	188.184.67.127	10.21.58.50	HTTP	932	HTTP/1.1 200 OK (text/html)
366	6.722605	10.21.58.50	188.184.67.127	HTTP	438	GET /favicon.ico HTTP/1.1
378	6.834141	188.184.67.127	10.21.58.50	HTTP	488	HTTP/1.1 200 OK (image/vnd.microsoft.icon)

> Frame 357: Packet, 497 bytes on wire (3976 bits), 497 bytes captured (3976 bits) on interface \Device\NPF_{...}	0000	3a a3 2b ce b3 84 12 a3 ef 62 d1 5b 08 00 45 00	...
> Ethernet II, Src: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b), Dst: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)	0010	01 e3 43 e1 40 00 00 06 70 b5 0a 15 3a 32 bc b8	...
> Internet Protocol Version 4, Src: 10.21.58.50, Dst: 188.184.67.127	0020	43 7f ca b7 00 50 e8 e4 a2 66 34 0c 29 58 50 18	...
> Transmission Control Protocol, Src Port: 50359, Dst Port: 80, Seq: 1, Ack: 1, Len: 443	0030	02 00 e3 cb 00 00 47 45 54 20 2f 20 48 54 54 50	...
Source Port: 50359	0040	2f 31 2e 31 0d 0a 4b 6f 73 74 3a 20 69 6e 66 6f	...
Destination Port: 80	0050	2e 63 65 72 6e 2e 63 68 0d 0a 43 6f 6e 6e 65 63	...
[Stream index: 1]	0060	74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65	...
[Stream Packet Number: 4]	0070	0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63 75	...
> [Conversation completeness: Complete, WITH_DATA (31)]	0080	72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 0d 0a	...
[TCP Segment Len: 443]	0090	55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69	...
Sequence Number: 1 (relative sequence number)	00a0	6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73	...
Sequence Number (raw): 3907297894	00b0	20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36 34 3b	...
[Next Sequence Number: 444 (relative sequence number)]	00c0	20 78 36 34 29 20 41 70 70 6c 65 57 65 62 4b 69	...
Acknowledgment Number: 1 (relative ack number)	00d0	74 2f 35 33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c	...
Acknowledgment number (raw): 873212248	00e0	20 6c 69 6b 65 20 47 65 63 6b 6f 29 20 43 68 72	...
0101 = Header Length: 20 bytes (5)	00f0	6f 6d 65 2f 31 35 30 2e 30 2e 30 2e 30 20 59 61	...
> Flags: 0x018 (PSH, ACK)	0100	42 72 6f 77 73 65 72 2f 32 36 2e 38 2e 30 2e 30	...
Window: 512	0110	20 53 61 66 61 72 69 2f 35 33 37 2e 33 36 0d 0a	...
[Calculated window size: 131072]	0120	41 63 63 65 70 74 3a 20 74 65 78 74 2f 68 74 6d	...
Checksum: 0xe3cb [unverified]	0130	6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e 2f 70 68	...
[Checksum Status: Unverified]	0140	74 6d 6c 2b 78 6d 6c 2c 61 70 70 6c 69 63 61 74	...
Urgent Pointer: 0	0150	69 6f 6e 2f 78 6d 6c 3b 71 3d 30 2e 39 2c 69 6d	...
> [Timestamps]	0160	61 67 65 2f 61 76 69 66 2c 69 6d 61 67 65 2f 77	...
> [SEQ/ACK analysis]	0170	65 62 70 2c 69 6d 61 67 65 2f 61 70 6e 67 2c 2a	...
[Client Contiguous Streams: 1]	0180	2f 2a 3b 71 3d 30 2e 38 2c 61 70 70 6c 69 63 61	...
[Server Contiguous Streams: 1]	0190	74 69 6f 6e 2f 73 69 67 6e 65 64 2d 65 78 63 68	...
TCP payload (443 bytes)	01a0	61 6e 67 65 3b 76 3d 62 33 3b 71 3d 30 2e 37 0d	...
> Hypertext Transfer Protocol	01b0	0a 41 63 63 65 70 74 2d 45 6e 63 6f 64 69 6e 67	...
	01c0	3a 20 67 7a 69 70 2c 20 64 65 66 6c 61 74 65 0d	...
	01d0	0a 41 63 63 65 70 74 2d 4c 61 6e 67 75 61 67 65	...
	01e0	3a 20 72 75 2c 65 6e 3b 71 3d 30 2e 39 0d 0a 0d	...
	01f0	0a	...

Рисунок 11: Анализ HTTP-запроса и TCP-заголовка в Wireshark

2.12 Исследуя HTTP-ответ

No.	Time	Source	Destination	Protocol	Length	Info
329	5.960346	10.21.58.50	188.184.67.127	HTTP	342	GET /apple-touch-icon.png HTTP/1.1
332	6.053248	188.184.67.127	10.21.58.50	HTTP	414	HTTP/1.1 404 Not Found (text/html)
357	6.532674	10.21.58.50	188.184.67.127	HTTP	497	GET / HTTP/1.1
359	6.634118	188.184.67.127	10.21.58.50	HTTP	932	HTTP/1.1 200 OK (text/html)
366	6.722695	10.21.58.50	188.184.67.127	HTTP	438	GET /favicon.ico HTTP/1.1
378	6.834141	188.184.67.127	10.21.58.50	HTTP	408	HTTP/1.1 200 OK (image/vnd.microsoft.icon)

> Frame 359: Packet, 932 bytes on wire (7456 bits), 932 bytes captured (7456 bits) on interface \Device\NPF...	0000	12 a3 ef 62 d1 5b 3a a3	2b ce b3 84 08 00 45 28	...	b [
> Ethernet II, Src: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84), Dst: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)	0010	03 96 c9 03 40 80 2d 06	3c 08 bc b8 43 7f 0a 15	...	0 [
> Internet Protocol Version 4, Src: 188.184.67.127, Dst: 10.21.58.50	0020	3a 32 00 50 c4 b7 34 0c	20 58 e8 e4 a4 21 58 18	12 p...	
> Transmission Control Protocol, Src Port: 80, Dst Port: 58359, Seq: 1, Ack: 444, Len: 878	0030	01 f5 42 5f 00 00 48 54	54 58 2f 31 2e 31 28 32	-B...	
Source Port: 80	0040	30 30 20 4f 4b 0d 0a 44	61 74 65 3a 20 53 61 74	00 OK	
Destination Port: 58359	0050	2c 20 32 36 20 53 65 70	20 32 30 32 36 20 31 30	, 26 5	
[Stream index: 1]	0060	3a 32 33 3a 35 36 20 47	4d 54 0d 0a 53 65 72 76	:23:56	
[Stream Packet Number: 0]	0070	65 72 3a 20 41 70 61 65	68 65 0d 0a 4c 61 73 74	err: Ap	
[Conversation completeness: Complete, WITH_DATA (31)]	0080	2d 4d 6f 64 69 66 69 65	64 3a 20 57 65 64 2c 20	-Modif	
[TCP Segment Len: 878]	0090	30 35 20 46 65 62 20 32	30 31 34 20 31 36 3a 30	05 Feb	
Sequence Number: 1 (relative sequence number)	00a0	30 3a 33 31 20 47 4d 54	8d 0a 45 54 61 67 3a 20	0:31 G	
Sequence Number (raw): 823212248	00b0	22 32 38 36 2d 34 66 31	61 61 64 62 33 31 30 35	"286-4	
[Next Sequence Number: 879 (relative sequence number)]	00c0	63 30 22 0d 0a 41 63 63	65 70 74 2d 52 61 6e 67	cb" A	
Acknowledgment Number: 444 (relative ack number)	00d0	65 73 3a 20 62 79 74 65	73 0d 0a 43 6f 6e 74 65	esi: by	
Acknowledgment Number (raw): 3907298337	00e0	66 74 2d 4c 65 6e 67 74	68 3a 20 36 34 36 0d 0a	nt-len	
Window: 581	00f0	43 6f 6e 66 65 63 74 69	6f 6e 3a 20 63 6c 6f 73	Connec	
[Calculated window size: 64128]	0100	65 0d 0a 43 6f 6e 74 65	6e 74 2d 54 79 70 65 3a	e Con	
[Window size scaling factor: 128]	0110	20 74 65 78 74 2f 68 74	6d 6c 0d 0a 0d 0a 3c 68	text/	
Checksum: 0x425f [unverified]	0120	74 6d 6c 3e 3c 68 65 61	64 3e 3c 2f 68 65 61 64	telch	
[Checksum Status: Unverified]	0130	3e 3c 62 6f 64 79 3e 3c	68 65 61 64 65 72 3e 0a	>body	
Urgent Pointer: 0	0140	3c 74 69 74 6c 65 3e 68	74 74 70 3a 2f 2f 69 6e	<title	
[Timestamps]	0150	66 6f 2e 63 65 72 6e 2e	63 68 3c 2f 74 69 74 6c	fo:cer	
[SEQ/ACK analysis]	0160	65 3e 0a 3c 2f 68 65 61	64 65 72 3e 0a 0a 3c 68	es-</h	
[Client Contiguous Streams: 1]	0170	31 3e 68 74 74 70 3a 2f	2f 69 6e 66 6f 2e 63 65	1>http	
[Server Contiguous Streams: 1]	0180	72 6e 2e 63 68 20 2d 20	68 6f 6d 65 20 6f 66 20	rn.ch	
TCP payload (878 bytes)	0190	74 68 65 20 66 69 72 73	74 20 77 65 62 73 69 74	the fi	
> Hypertext Transfer Protocol	01a0	65 3c 2f 68 31 3e 0a 3c	70 3e 46 72 6f 6d 20 68	e</h1>	
> Line-based text data: text/html (13 lines)	01b0	65 72 65 20 79 6f 75 20	63 61 6e 3a 3c 2f 70 3e	ere yo	
	01c0	0a 3c 75 6c 3e 0a 3c 6c	69 3e 61 20 68 72 65	-cul>	
	01d0	66 3d 22 68 74 74 70 3a	2f 2f 69 6e 66 6f 2e 63	fe="htt	
	01e0	65 72 6e 2e 63 68 2f 68	79 70 65 72 74 65 78 74	ern.ch	
	01f0	2f 57 57 57 2f 54 68 65	50 72 6f 6a 65 63 74 2e	/html/	
	0200	68 74 6d 6c 22 3e 42 72	6f 77 73 65 20 74 68 65	html">	
	0210	20 66 69 72 73 74 20 77	65 62 73 69 74 65 3c 2f	first	
	0220	61 3e 3c 2f 6c 69 3e 0a	3c 6c 69 3e 3c 61 20 68	a</li	
	0230	72 65 66 3d 22 68 74 74	70 3a 2f 2f 6c 69 6e 65	ref="h	
	0240	2d 6d 6f 64 65 2e 63 65	72 6e 3e 63 68 2f 77 77	-mode	
	0250	77 2f 68 79 70 65 72 74	65 78 74 2f 57 57 57 2f	w/hype	
	0260	54 68 65 50 72 6f 6a 65	63 74 2e 68 74 6d 6c 22	ThePro	
	0270	3e 42 72 6f 77 73 65 20	74 68 65 20 66 69 72 73	>Brows	
	0280	74 20 77 65 62 73 69 74	65 20 75 73 69 6e 67 20	t webs	
	0290	74 6e 6c 3e 6c 6e 6e 6e	74 64 64 64 6e 6e 6e 73	the 14	

Рисунок 12: Анализ HTTP-ответа 200 OK и TCP-заголовка

2.13 Анализирую DNS-запрос

No.	Time	Source	Destination	Protocol	Length	Info
338	6.087050	10.21.58.106	10.21.58.50	DNS	129	Standard query response 0xf340 HTTPS favicon.yandex.net SOA ns1.yandex.net
339	6.095626	10.21.58.106	10.21.58.50	DNS	142	Standard query response 0x1f0a A favicon.yandex.net A 213.180.204.36 A 77.88.21.36 A 87.250.251.13
393	8.517572	10.21.58.50	10.21.58.106	DNS	76	Standard query 0xe158 HTTPS home.web.cern.ch
394	8.517888	10.21.58.50	10.21.58.106	DNS	76	Standard query 0xdd46 A home.web.cern.ch
395	8.740602	10.21.58.50	10.21.58.106	DNS	76	Standard query 0x42a9 HTTPS home.web.cern.ch
396	8.740600	10.21.58.50	10.21.58.106	DNS	76	Standard query 0x8bf7 A home.web.cern.ch
397	8.764733	10.21.58.106	10.21.58.50	DNS	174	Standard query response 0xdd46 A home.web.cern.ch CNAME app-catalogue-apps-lb-home.cern.ch A 10.21.58.106
398	8.772230	10.21.58.106	10.21.58.50	DNS	183	Standard query response 0xe158 HTTPS home.web.cern.ch CNAME app-catalogue-apps-lb-home.cern.ch
399	8.772230	10.21.58.106	10.21.58.50	DNS	183	Standard query response 0x42a9 HTTPS home.web.cern.ch CNAME app-catalogue-apps-lb-home.cern.ch

> Frame 393: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface \Device\NPF_{234A...}	0000	3a a3 2b ce b3 04 12 a3 ef 62 d1 5b 00 00 45 00	:>...
> Ethernet II, Src: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b), Dst: 3a:a3:2b:ce:b3:04 (3a:a3:2b:ce:b3:04)	0010	00 3e 9b eb 00 00 00 11 15 fe 0a 15 3a 32 0a 15	:>...>
> Internet Protocol Version 4, Src: 10.21.58.50, Dst: 10.21.58.106	0020	3a 6a d6 da 00 35 00 2a ca ae e1 58 01 00 00 01	:j>...
> User Datagram Protocol, Src Port: 55002, Dst Port: 53	0030	00 00 00 00 00 00 04 68 6f 6d 65 03 77 65 62 04	:>...>
Source Port: 55002	0040	63 65 72 6e 02 63 68 00 00 41 00 01	cern-
Destination Port: 53			
Length: 42			
Checksum: 0xcaae [unverified]			
[Checksum Status: Unverified]			
[Stream index: 28]			
[Stream Packet Number: 1]			
> [Timestamps]			
UDP payload (34 bytes)			
> Domain Name System (query)			
Transaction ID: 0xe158			
> Flags: 0x0100 Standard query			
Questions: 1			
Answer RRs: 0			
Authority RRs: 0			
Additional RRs: 0			
> Queries			
[Response In: 398]			

Рисунок 13: Анализ DNS-запроса, передаваемого по UDP

2.14 Анализирую DNS-ответ

The screenshot displays the Wireshark network protocol analyzer interface. The top pane shows a list of captured packets, with packet 398 selected. The middle pane shows the details of the selected packet, and the bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
338	6.087050	10.21.58.106	10.21.58.50	DNS	129	Standard query response 0xf340 HTTPS favicon.yandex.net SOA ns1.yandex.net
339	6.095626	10.21.58.106	10.21.58.50	DNS	142	Standard query response 0x1f0a A favicon.yandex.net A 213.180.204.36 A 77.88.21.36 A 87.250.251.100
393	8.517572	10.21.58.50	10.21.58.106	DNS	76	Standard query 0xe158 HTTPS home.web.cern.ch
394	8.517888	10.21.58.50	10.21.58.106	DNS	76	Standard query 0xdd46 A home.web.cern.ch
395	8.740602	10.21.58.50	10.21.58.106	DNS	76	Standard query 0x42a9 HTTPS home.web.cern.ch
396	8.740680	10.21.58.50	10.21.58.106	DNS	76	Standard query 0x8bf7 A home.web.cern.ch
397	8.764733	10.21.58.106	10.21.58.50	DNS	174	Standard query response 0xdd46 A home.web.cern.ch CNAME app-catalogue-apps-lb-home.cern.ch A 10.21.58.106
398	8.772230	10.21.58.106	10.21.58.50	DNS	183	Standard query response 0xe158 HTTPS home.web.cern.ch CNAME app-catalogue-apps-lb-home.cern.ch
399	8.772230	10.21.58.106	10.21.58.50	DNS	183	Standard query response 0x42a9 HTTPS home.web.cern.ch CNAME app-catalogue-apps-lb-home.cern.ch

Packet 398 Details:

- Frame 398: Packet, 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits) on interface \Device\NPF...
- Ethernet II, Src: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84), Dst: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b)
- Internet Protocol Version 4, Src: 10.21.58.106, Dst: 10.21.58.50
- User Datagram Protocol, Src Port: 53, Dst Port: 55002
 - Source Port: 53
 - Destination Port: 55002
 - Length: 149
 - Checksum: 0x67b9 [unverified]
 - [Checksum Status: Unverified]
 - [Stream Index: 28]
 - [Stream Packet Number: 2]
 - [Timestamps]
 - UDP payload (141 bytes)
- Domain Name System (response)
 - Transaction ID: 0xe158
 - Flags: 0x8180 Standard query response, No error
 - Questions: 1
 - Answer RRs: 1
 - Authority RRs: 1
 - Additional RRs: 0
 - Queries
 - Answers
 - Authoritative nameservers
 - Request In: 393
 - Time: 254.658000 milliseconds

Raw Data (Hex):

```
0000 12 a3 ef 62 d1 5b 3a a3 2b ce b3 84 00 00 45 00 ...b-
0010 00 a9 6e 41 40 00 00 11 43 3d 0a 15 3a 6a 0a 15 ...na@
0020 3a 32 00 35 d6 da 00 95 67 b9 e1 58 01 00 00 01 :2.5
0030 00 01 00 01 00 00 04 68 6f 6d 65 03 77 65 62 04 ....
0040 63 65 72 6e 02 63 68 00 00 41 00 01 c0 0c 00 05 cern-
0050 00 01 00 00 00 3c 00 24 1a 61 70 70 2d 63 61 74 ....
0060 61 6c 6f 67 75 65 2d 61 70 70 73 2d 6c 62 2d 68 alogu
0070 6f 6d 65 04 63 65 72 6e 02 63 68 00 c0 2e 00 06 ome:c
0080 00 01 00 00 00 3c 00 2f 00 65 78 74 2d 64 6e 73 ....
0090 2d 31 c0 49 0c 65 78 74 65 72 6e 61 6c 2d 64 6e -1 I-
00a0 73 c0 49 00 00 00 21 00 00 0e 10 00 00 01 2c 00 s I-
00b0 36 ee 00 00 00 00 3c 6-----
```

Рисунок 14: Анализ DNS-ответа, передаваемого по UDP

2.15 Проверяю наличие QUIC-трафика

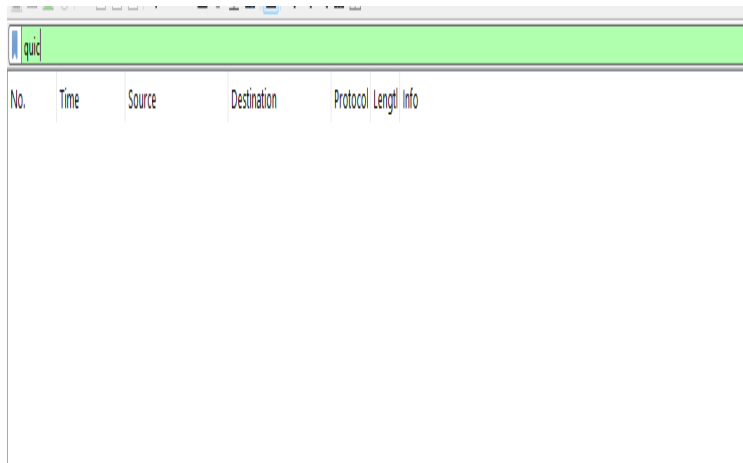


Рисунок 15: Результат применения фильтра quic в Wireshark

2.16 Анализирую TLS-трафик

The image shows a Wireshark packet capture of a TLS handshake. The packet list on the left shows a sequence of packets from 24 to 48. The packet details pane on the right shows the structure of the selected packet (packet 24, a Client Hello). The packet bytes pane on the right shows the raw data of the selected packet.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
24	0.242579	10.21.58.50	87.250.251.183	TLSv1.2	781	Client Hello (SHA256, browser.yandex.net)
28	0.284134	10.21.58.50	188.184.67.127	TLSv1.3	578	Client Hello (SHA256, info.cern.ch)
34	0.387593	87.250.251.183	10.21.58.50	TLSv1.2	210	Server Hello, Change Cipher Spec, Encrypted Handshake Message
35	0.387865	10.21.58.50	87.250.251.183	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
36	0.388837	10.21.58.50	87.250.251.183	TLSv1.2	1167	Application Data
37	0.388873	10.21.58.50	87.250.251.183	TLSv1.2	1814	Application Data
39	0.429599	188.184.67.127	10.21.58.50	TLSv1.3	1354	Server Hello, Change Cipher Spec, Application Data
46	0.433634	188.184.67.127	10.21.58.50	TLSv1.3	746	Application Data, Application Data, Application Data
48	0.435470	10.21.58.50	188.184.67.127	TLSv1.3	134	Change Cipher Spec, Application Data

Packet Details (Packet 24):

- Ethernet II, Src: 12:a3:ef:62:d1:5b (12:a3:ef:62:d1:5b), Dst: 3a:a3:2b:ce:b3:84 (3a:a3:2b:ce:b3:84)
- Internet Protocol Version 4, Src: 10.21.58.50, Dst: 188.184.67.127
- Transmission Control Protocol, Src Port: 62156, Dst Port: 443, Seq: 1301, Ack: 1, Len: 524
 - Source Port: 62156
 - Destination Port: 443
 - [Stream index: 2]
 - [Stream Packet Number: 5]
 - [Conversation completeness: Complete, WITH_DATA (31)]
 - [TCP Segment Len: 524]
 - Sequence Number: 1301 (relative sequence number)
 - Sequence Number (raw): 2422384355
 - [Next Sequence Number: 1825 (relative sequence number)]
 - Acknowledgment Number: 1 (relative ack number)
 - Acknowledgment number (raw): 202571935
 - 0101 = Header Length: 20 bytes (5)
 - Flags: 0x018 (PSH, ACK)
 - Window: 512
 - [Calculated window size: 131072]
 - [Window size scaling factor: 256]
 - Checksum: 0xb012 [unverified]
 - Checksum Status: Unverified
 - Urgent Pointer: 0
- [Timestamps]
- [SEQ/ACK analysis]
- [Client Contiguous Streams: 1]
- [Server Contiguous Streams: 1]
- TCP payload (524 bytes)
- TCP segment data (524 bytes)
- [2 Reassembled TCP Segments (1824 bytes): #27(1300), #28(524)]
- Transport Layer Security
 - [Stream index: 1]
 - TLSv1.3 Record Layer: Handshake Protocol: Client Hello
 - Content Type: Handshake (22)
 - Version: TLS 1.0 (0x0301)
 - Length: 1819
 - Handshake Protocol: Client Hello

Packet Bytes (Packet 24):

0000 3a a3 2b ce b3 84 12 a3 ef 62 d1 5b 08 00 45 00 1:--
0010 02 34 43 c7 40 00 80 06 70 7e 0a 15 3a 32 bc b8 :4C@
0020 43 7f f2 cc 01 bb 90 62 a6 e3 78 be 05 5f 50 18 C:--
0030 02 00 00 12 00 00 66 56 05 a0 18 54 9c 81 22 ad :---
0040 3e a1 1d 7d 3b 3f a1 11 2d 2e 06 cb 00 0b 12 59 :>:~
0050 d0 7f 92 53 ab 3d 94 63 d9 32 c7 a7 c7 34 42 15 :--S-
0060 4c 7e b4 b6 de 40 78 36 a6 65 4b 15 1c 5d e2 2a L:--
0070 32 44 40 2f e9 45 b6 80 5b ee 1b ab 08 53 5c 07 200/1
0080 62 04 b0 b8 08 c9 98 32 c5 44 90 33 40 6d 0e 3c b:--
0090 86 97 0b 3d fa eb 1a 45 1a 78 1a c3 5c 97 e1 88 :---
00a0 2e 04 36 3a b9 23 a1 73 56 ce bb b5 9e 44 4b c2 :.6:4
00b0 88 a7 0d d1 71 6e 53 7f c7 a7 77 9b 62 a0 fa 95 :--q
00c0 36 fd c4 3e 53 a9 53 c2 72 68 6a 13 75 12 39 ad 6:--P
00d0 76 c3 69 2a e3 4c 90 3a cb a9 ba 34 dd 83 85 47 v:~1
00e0 13 3b bd 95 7f 08 fb 62 a0 68 26 70 c2 51 e2 c1 :>:~
00f0 3a c7 d6 6a f6 41 bf 5f e0 ae f8 6b 34 3f f5 70 :>:f
0100 02 94 2c 8f a5 91 b8 5c 47 1b 54 1b 08 d6 76 1f :>:f
0110 00 0e 5c 46 b9 ed d0 08 ec 5a 58 64 d1 08 44 33 :>:V
0120 b9 c4 48 88 97 6a cd b3 b1 31 29 22 9e 09 c7 80 :>:H
0130 0a b5 0b 67 d2 04 fb 00 29 a6 1c 6d 09 42 ae 44 :>:~
0140 80 40 74 c3 a6 c7 93 44 a8 75 3b ff 34 9d 99 07 :>:~
0150 36 4d 12 48 ee 18 54 90 49 45 42 50 2c 6c 99 0f 6M:H
0160 b4 58 9f 42 ab 34 85 e2 90 b1 07 51 f9 cb a7 2f :X:B
0170 22 07 fb 28 04 07 a0 6e c5 7c a4 80 f2 9d 53 ab :>:~
0180 26 ef 39 c5 9e 23 c2 bd 86 62 99 24 66 21 5f 71 :>:B
0190 7a 70 9a 99 c4 93 a0 b5 9f ee 77 8c 00 3c 1a 33 :>:p
01a0 54 95 e3 93 5b 73 90 77 3c a4 49 de 3d 01 15 6d T:~[
01b0 bd 04 55 49 5b 8a 19 40 72 85 1c 6d 46 52 ce 0b :>:UT
01c0 1b 5d 6d 4e 76 d7 60 09 71 91 28 c0 22 ae 99 e2 :>:m
01d0 35 dd 46 00 af 3f 14 e4 53 68 ce 91 b4 df 6d 03 5:~
01e0 3a 7d dc 2e 98 72 ee 0b 2b d5 73 00 1d 00 20 c9 :>:~
01f0 bc 3c 49 d7 9d d3 fb 9e 78 74 e7 a6 eb 5f 34 :>:C
0200 31 11 2d 82 55 ab 5f 6a 4c dd 3a d8 4d fa 2d 10 :>:U
0210 23 00 00 00 12 00 00 00 17 00 00 00 10 00 00 00 :>:~
0220 0c 02 68 32 08 68 74 70 2f 31 2e 31 00 05 00 :>:h
0230 05 01 00 00 00 00 00 1b 00 03 02 00 02 2a 00 :>:~
0240 01 00 :>:~

Рисунок 16: Анализ TLS-трафика при отсутствии пакетов QUIC

2.17 Изучаю TCP handshake

tcp

No.	Time	Source	Destination	Protocol	Length	Info
9	0.097390	87.240.137.130	10.21.58.50	TCP	54	443 → 61448 [ACK] Seq=484 Ack=260 Win=21 Len=0
14	1.323840	10.21.58.50	188.184.67.127	TCP	66	51410 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
15	1.327604	10.21.58.50	188.184.67.127	TCP	66	52119 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
16	1.415947	188.184.67.127	10.21.58.50	TCP	66	80 → 51410 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1300 SACK_PERM WS=128
17	1.416115	10.21.58.50	188.184.67.127	TCP	54	51410 → 80 [ACK] Seq=1 Ack=1 Win=131072 Len=0
18	1.416918	10.21.58.50	188.184.67.127	HTTP	609	GET / HTTP/1.1
19	1.417221	188.184.67.127	10.21.58.50	TCP	66	80 → 52119 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1300 SACK_PERM WS=128
20	1.417303	10.21.58.50	188.184.67.127	TCP	54	52119 → 80 [ACK] Seq=1 Ack=1 Win=131072 Len=0
21	1.406341	188.184.67.127	10.21.58.50	TCP	54	80 → 51410 [ACK] Seq=1 Ack=556 Win=63744 Len=0
22	1.408327	188.184.67.127	10.21.58.50	HTTP	250	HTTP/1.1 304 Not Modified
23	1.408844	10.21.58.50	188.184.67.127	TCP	54	51410 → 80 [FIN, ACK] Seq=556 Ack=197 Win=131072 Len=0
24	1.500113	188.184.67.127	10.21.58.50	TCP	54	80 → 51410 [FIN, ACK] Seq=197 Ack=556 Win=63744 Len=0
25	1.500149	10.21.58.50	188.184.67.127	TCP	54	51410 → 80 [ACK] Seq=557 Ack=198 Win=131072 Len=0
28	1.515435	10.21.58.50	87.250.251.20	TLSv1.2	169	Application Data

[Stream Index: 1]
[Stream Packet Number: 1]
> [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 4288130533
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1000 = Header Length: 32 bytes (0)
▼ [Flags: 0x002 (SYN)]
0000 = Reserved: Not set
...0 = Accurate ECN: Not set
...0 = Congestion Window Reduced: Not set
...0 = ECN-Echo: Not set
...0 = Urgent: Not set
...0 = Acknowledgment: Not set
...0 = Push: Not set
...0 = Reset: Not set
> [.... .0.0. = Syn: Set
...0 = Fin: Not set
[TCP Flags:5.]
Window: 64240
[calculated window size: 64240]
Checksum: 0xb900 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
> Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Op
> [Timestamps]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]

0000 3a a3 2b ce b5 84 12 a3 ef 62 d1 5b 00 00 45 00 :+...
0010 00 34 43 e8 40 00 00 06 72 5d 0a 15 3a 32 bc b8 :4C@
0020 43 7f c8 d2 00 50 ff 97 ad e5 00 00 00 00 02 C...P
0030 fa f0 b9 00 00 00 02 04 05 b4 01 03 03 00 01 01 :...
0040 04 02

Рисунок 17: Анализ трёхэтапного TCP handshake в Wireshark

2.18 Строю график потока TCP

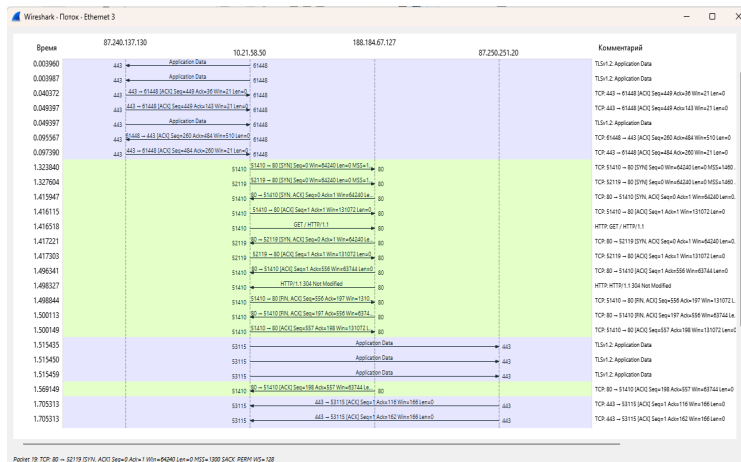


Рисунок 18: График потока TCP-соединений и анализ TCP handshake

3. 3. Выводы по проделанной работе

В ходе лабораторной работы была изучена сетевая конфигурация компьютера, определены IP- и MAC-адреса сетевых интерфейсов, а также рассмотрена структура MAC-адресов.

С помощью Wireshark был выполнен захват и анализ кадров ARP и ICMP, исследованы HTTP- и DNS-запросы, особенности передачи данных по TCP и UDP, а также рассмотрен процесс установления TCP-соединения через последовательность **SYN → SYN, ACK → ACK**.